Gold Future/ETF Analysis

Joe Hollander

2025

Maturity	**Slope**	**Intercept**	R^2	**RMSE**
1-Month	0.99962	1.2811×10^{-7}	0.94937	0.00221
2-Month	1.00241	-1.3769×10^{-6}	0.94710	0.00227
6-Month	1.00322	8.1922×10^{-7}	0.94068	0.00242
12-Month	0.94779	2.2525×10^{-5}	0.83986	0.00409

Table 1: New Data: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

The slopes for the 1, 2, and 6-month contracts are all very close to 1, indicating a strong correlation with spot gold returns. The 12-month contract, however, has a noticeably lower slope of 0.94779 and a significantly lower R^2 of 0.83986. This may be a result of the lower liquidity and trading volume of the longer-dated contract, making it less sensitive to daily spot price movements. The great differnce in all statistics, though, could also indicate that the data for the 12-month contract data could be corrupted, recorded incorrectly, or otherwise flawed.

Maturity	**Slope**	**Intercept**	R^2	**RMSE**
1-Month	0.94314	2.4191×10^{-5}	0.80947	0.00530
2-Month	0.94301	1.0000×10^{-5}	0.80911	0.00530
6-Month	0.94348	3.1797×10^{-5}	0.80934	0.00530
12-Month	0.94358	-5.2333×10^{-5}	0.80984	0.00529

Table 2: Leung & Ward (2015) Data: Regression coefficients for one-day futures returns on one-day spot gold returns.

The slopes are also very close to 1, but consistently lower than the newer data. The R^2 values are also significantly lower than the newer data. This may be due to the smaller sample size of the older data and lower liquidity. However, the much greater consistency of the Leung & Ward slopes and the lower R^2 values could also indicate a procedural difference in how the statistics were calculated.

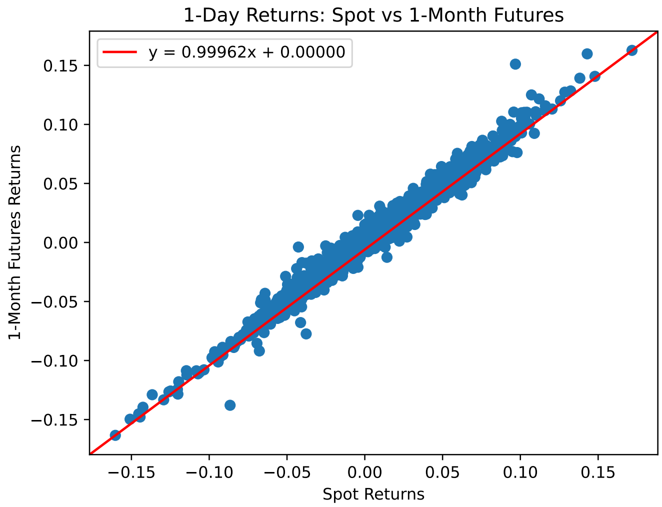

Figure 1: Scatter plot of one-day returns for spot gold versus 1-month gold futures.

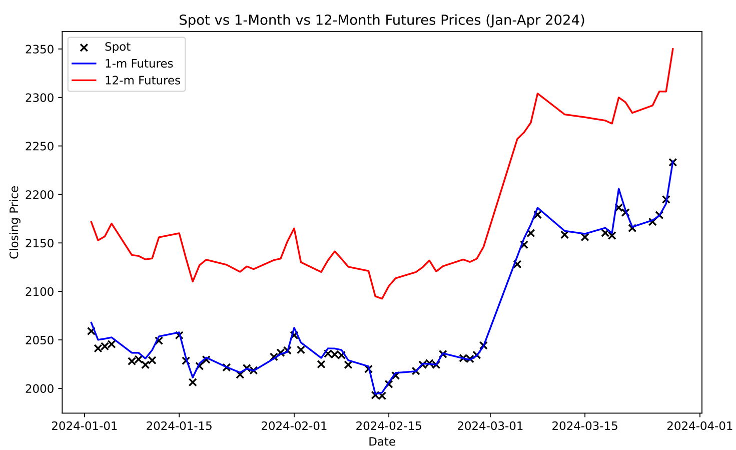

Figure 2: Comparison of closing prices for spot gold, 1-month futures, and 12-month futures in early 2024. The 1-month contract tracks the spot price closely, while the 12-month contract maintains a significant premium.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.99962	1.00241	1.00322	0.94779
	5	1.00741	1.01021	1.01058	0.98753
	10	1.00224	1.00399	1.00337	0.98967
	15	1.00312	1.00494	1.00331	0.99205
R^2	1	0.94937	0.94710	0.94068	0.83986
	5	0.98419	0.98398	0.98117	0.95670
	10	0.99152	0.99101	0.98776	0.97473
	15	0.99386	0.99330	0.98994	0.98068

Table 3: New Data: Slopes and R^2 from the regressions of futures returns versus gold returns over different holding periods.

The slopes and R^2 values generally increase with longer holding periods but decrease with longer-dated contracts. This suggests that futures returns are correlated with spot returns over longer horizons, with shorter measurement periods capturing more noise.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.94314	0.94301	0.94348	0.94358
	5	0.99805	0.99752	0.99757	0.99715
	10	1.01075	1.01020	1.00970	1.00841
	15	0.99448	0.99461	0.99431	0.99345
R^2	1	0.80947	0.80911	0.80934	0.80984
	5	0.97154	0.97158	0.97194	0.97165
	10	0.98516	0.98530	0.98572	0.98550
	15	0.98704	0.98691	0.98725	0.98713

Table 4: Leung & Ward (2015) Data: Slopes and R^2 from regressions of futures returns vs. gold returns over different holding periods.

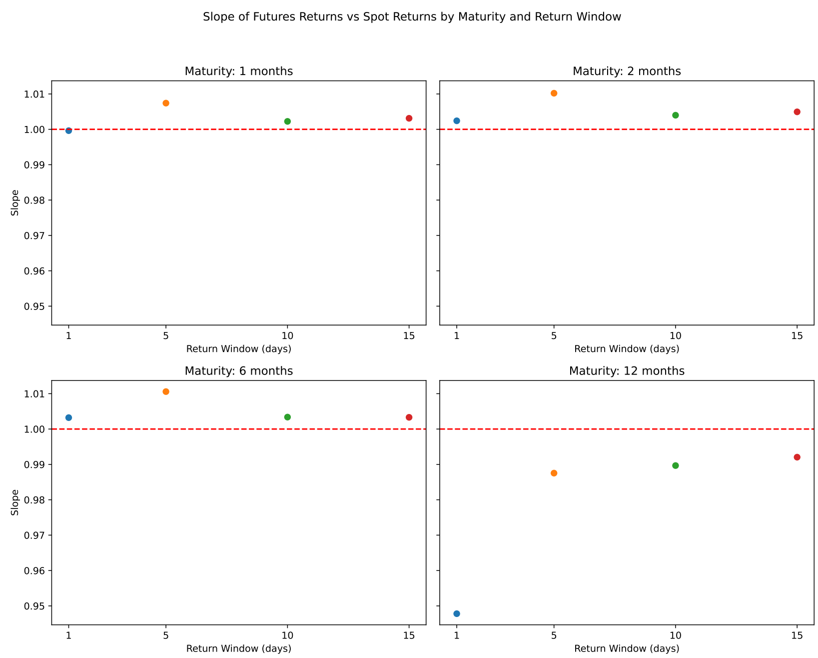

Figure 3: Regression slope of futures vs. spot returns, analyzed by contract maturity and the return window length in days. For most maturities, the slope approaches 1 as the return window lengthens, but the 12-month contract is still significantly different from 1 compared to the other maturities.

The out of sample period is defined as starting on January 1, 2024 and is approximately 10% of the data. The RMSEs are calculated in dollar terms assuming a notional investment of $1000 in the spot gold price. The weights are calculated using a constrained least squares regression that forces the weights to sum to one.

Portfolio	w_0	w_1	w_2	w_6	w_{12}	$\mathbf{RMSE_{in}}$	$\mathbf{RMSE_{out}}$
1-m	0.00258	0.99741	—	—	—	$2.67	$5.12
2-m	0.00679	—	0.99322	—	—	$3.70	$5.69
6-m	0.02215	—	—	0.97785	—	$10.07	$13.34
12-m	0.04517	—	—	—	0.95483	$18.80	$17.72
1-m, 2-m	0.00170	1.21350	-0.21521	—	—	$2.62	$5.03
1-m, 6-m	0.00168	1.04713	—	-0.04881	—	$2.63	$4.86
1-m, 12-m	0.00192	1.01384	—	—	-0.01576	$2.68	$4.91
2-m, 6-m	0.00333	—	1.23020	-0.23352	—	$3.21	$4.62
2-m, 12-m	0.00311	—	1.09427	—	-0.09738	$3.29	$4.94
6-m, 12-m	0.00444	—	—	1.76141	-0.76585	$6.08	$11.87
1-m, 2-m, 6-m	0.00159	1.14787	-0.12240	-0.02706	—	$2.62	$4.93
1-m, 2-m, 12-m	0.00173	1.24621	-0.25302	—	0.00508	$2.66	$4.96
1-m, 6-m, 12-m	0.00201	1.10693	—	-0.17570	0.06676	$2.64	$4.52
2-m, 6-m, 12-m	0.00329	—	1.22206	-0.22099	-0.00436	$3.24	$4.43
1-m, 2-m, 6-m, 12-m	0.00190	1.22259	-0.14302	-0.14612	0.06465	$2.63	$4.60

Table 5: New Data: Weights and in/out of sample RMSEs for portfolios of 1, 2, 3, and 4 futures contracts.

The results show that using a single futures contract provides a reasonable replication, but performance diminishes as the maturity of the contract increases (out-of-sample RMSE rises from $5.12 to $17.72). Adding a second futures contract generally improves the out-of-sample RMSE, with the 2-month and 6-month combination performing best at $4.62. The best-performing portfolio overall is the three-contract combination of 2-month, 6-month, and 12-month futures, which achieved an out-of-sample RMSE of $4.43. Notably, several of these futures portfolios outperform the GLD ETF, which has an out-of-sample RMSE of $6.00 over the same period.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed non risus. Suspendisse lectus tortor, dignissim sit amet, adipiscing nec, ultricies sed, dolor. Cras elementum ultrices diam. Maecenas ligula massa, varius a, semper congue, euismod non, mi.

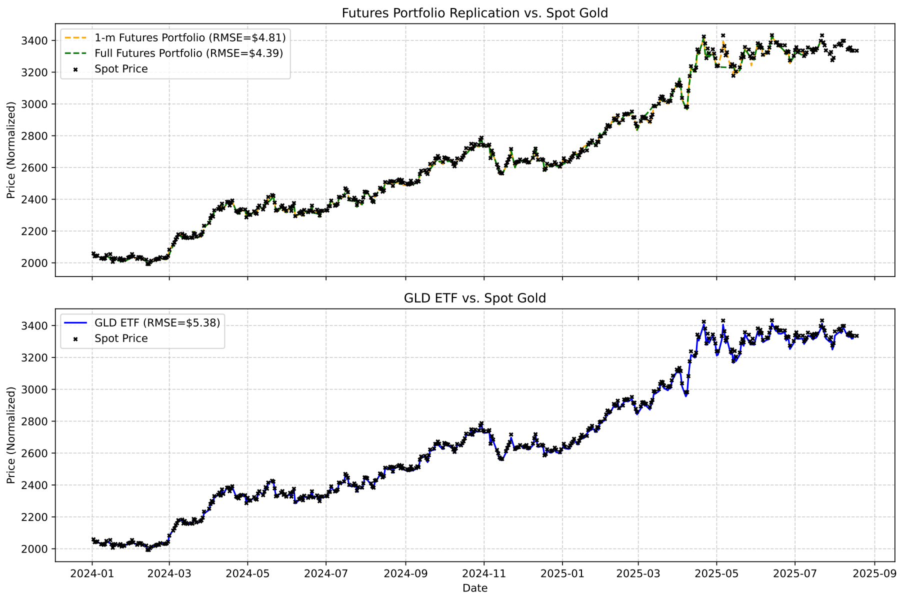

Figure 4: Visual comparison of tracking performance in early 2024. The top panel shows replication of spot gold using futures portfolios, while the bottom panel shows the GLD ETF's tracking of spot gold. The futures portfolios achieve a lower RMSE in this period than GLD.

Portfolio	w_0	w_{short}	w_{long}	$\textbf{RMSE}_{\textbf{in}}$	$\textbf{RMSE}_{\textbf{out}}$
1-m	-0.01071	1.01071	—	\$6.63	\$3.34
2-m	-0.04835	1.04835	—	\$12.22	\$6.12
12-m	-0.06842	1.06842	—	\$15.11	\$5.71
1-m, 2-m	-0.00088	1.27315	-0.27227	\$6.27	\$2.79
1-m, 6-m	-0.00171	1.23899	-0.23729	\$6.27	\$2.98
1-m, 12-m	-0.00021	1.19336	-0.19315	\$6.29	\$3.00
2-m, 6-m	-0.04079	5.06602	-4.02523	\$10.27	\$9.37

Table 6: Leung & Ward (2015) Data: Weights and in/out of sample RMSEs for portfolios of 1 and 2 futures contracts.

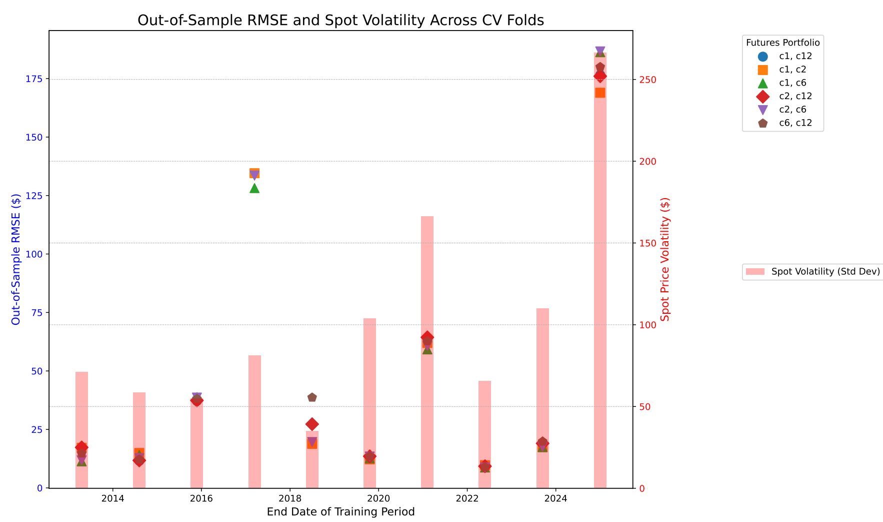

Figure 5: Out-of-sample RMSE for various futures portfolios across different cross-validation training periods. The pink bars represent the spot price volatility during each out-of-sample period.

(L)ETF	Leverage (β)	Slope	Intercept	t-stat	p-value	R^2	RMSE
GLD	1	1.00431	-1.30×10^{-5}	2.039	0.042	0.9780	0.00164
UGL	2	2.01860	-2.10×10^{-4}	2.615	0.009	0.9688	0.00326
GLL	-2	-2.01174	1.96×10^{-4}	-1.663	0.097	0.9690	0.00323
DGLD	-2	-1.97424	9.85×10^{-5}	0.499	0.618	0.4156	0.02058
UGLDF	3	3.00291	4.25×10^{-4}	0.026	0.979	0.2501	0.04519

Table 7: New Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

The regression results indicate that GLD, UGL, and GLL are effective at tracking their underlying benchmarks, with high R^2 values and slopes very close to their target betas. However, the low p-values for GLD (0.042) and UGL (0.009) suggest that their slopes are statistically different from the target betas of 1 and -1, respectively. DGLD and UGLDF perform poorly, with very low R^2 values (0.4156 and 0.2501), indicating they do not reliably track their leveraged benchmarks.

(L)ETF	Leverage (β)	Slope	Intercept	t-stat	p-value	R^2	RMSE
GLD	1	1.00540	-1.6427×10^{-6}	1.42692	0.15382	0.98060	0.00163
UGL	2	2.00572	-1.3107×10^{-4}	0.73319	0.46351	0.96257	0.00350
GLL	-2	-2.00609	1.2541×10^{-4}	-0.76019	0.44723	0.96317	0.00346
DGLD	-2	-1.97934	-2.0543×10^{-4}	0.40822	0.68316	0.43577	0.02026
UGLDF	3	3.01189	-1.9048×10^{-4}	0.11181	0.91099	0.28014	0.04369

Table 8: Leung & Ward (2015) Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

	$\text{RMSE}_{\text{diff}}$				Actual ETF RMSE
	1-Month	2-Month	6-Month	12-Month	
GLD	$113.94	$117.11	$120.92	$288.55	$55.79
UGL	$816.20	$825.18	$861.66	$907.40	$176.72
GLL	$1120.26	$1120.72	$1124.66	$1050.30	$1094.66
DGLD	$805.47	$805.11	$809.21	$773.20	$896.73
UGDF	$669.66	$644.20	$669.53	$1196.78	$198.11

Table 9: New Data: Dynamic replication one-day return differences ($\text{RMSE}_{\text{diff}}$) and (L)ETF RMSEs.

The performance of dynamic replication, which involves daily rebalancing of a single futures contract, varies significantly across different (L)ETFs. For positively leveraged ETFs like UGL (2x) and UGLDF (3x), as well as the inverse

ETF GLL (-2x), the $\text{RMSE}_{\text{diff}}$ for dynamic replication is substantially higher than the actual tracking error (Actual ETF RMSE) of the products themselves. For instance, the best dynamic replication for UGL (1-month contract) has an RMSE difference of \$816.20, whereas the ETF's actual RMSE is only \$176.72. This suggests that these products employ a more sophisticated strategy than simple daily rebalancing of a single futures contract. Conversely, for the inverse ETF DGLD (-2x), the dynamic replication RMSEs (e.g., \$805.47 for 1-month) are lower than the ETF's actual RMSE (\$896.73), implying that a simple dynamic strategy could have outperformed the product's own tracking.